Multimedia Signal Processing - 2nd Module

February, 18 2020

1. **Question (8 pts)**
   Consider a continuous source X that emits symbols following a probability distribution function (p.d.f.) $p_X(x)$.
   (a) Explain what is a Lloyd-Max quantizer and how it works.
   (b) Consider the following series of symbols emitted by X

   $$x = \{-1.6,\ 0.2,\ -1.6,\ -1.4,\ 1.4,\ -1.4,\ -0.4,\ 0.2,\ 1.6,\ -0.4,\ 2.4\}.$$

   Compute one full Lloyd-Max iteration considering $M = 3$ bins and the thresholds $\{-\infty, -1, 1, +\infty\}$ as initialization.
   (c) How many more iterations are needed to reach Lloyd-Max convergence.
   (d) Compute the quantized sequence $\tilde{x}$ applying the defined Lloyd-Max quantizer to x.
   (e) Which is the theoretical minimum number of bits per symbol needed to compress $\tilde{x}$ in a lossless way? (hint: estimate the p.m.f of the source from the observed $\tilde{x}$).

2. **Question (5 pts)**
   You want to encode an audio signal using a closed-loop DPCM.
   (a) Sketch the block-diagram of the encoder.
   (b) Sketch the block-diagram of the decoder.
   (c) Comment the sketched diagrams clearly explaining each block input-output relation.
   (d) Report one example in which this DPCM can be useful.

3. **Question (5 pts)**
   Consider two discrete memoryless sources X and Y characterized by a finite-element alphabet. You know the value of the entropy $H(X)$, the entropy $H(Y)$, and the mutual information $I(X;Y)$.
   (a) Using entropy properties, derive the value of the joint entropy $H(X;Y)$.
   (b) Using entropy properties, derive the value of the conditional entropy $H(X|Y)$.
   (c) What is the meaning of zero mutual information? In which case does it happen?

4. **Questions (5 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] In a typical video coding scheme, the decoder estimates motion vectors from the frames.
   - [T] [F] JPEG image coding standard makes use of 2D DCT.
   - [T] [F] Transform coding provides always better performance than a DPCM.
   - [T] [F] A discrete memoryless source X characterized by a finite-element alphabet can be encoded with less than $H(X)$ bit per symbol.
   - [T] [F] Masking effects can be used for efficiently allocating bits in perceptual audio coding schemes.

5. **Question (Matlab - 10pts)**

   Consider a uniformly distributed sequence of 1000 symbols `x_u = 100*rand(1, 1000)` and a gaussian distributed sequence of 1000 symbols `x_g = 50 + 100*randn(1, 1000)`.

   (a) Display the histogram of the two sequences.

   (b) Display the minimum and maximum value of both `x_u` and `x_g`.

   (c) For each sequence (i.e., `x_u` and `x_g`):

      i. Quantize the sequence using a uniform scalar quantizer with all the following quantization steps: $Q = \{2, 4, 6, 8, 10\}$.

      ii. For each quantization step Q, compute the MSE between the quantized and the original sequence.

      iii. For each quantization step Q, compute the entropy of the quantized sequence.

   (d) Plot the rate-distortion curve for each sequence (use entropy as rate and MSE as distortion).

   (e) Observing the rate-distortion curve, which sequence provides less distortion for the same rate? Why?